

Architecture part 1

Email → Classify and Route the question type → Critical keyword chat using fixed rules → Deepseek meaning check → Human review (critical==Y) or Specialist subagent(critical==N)
N → PDF evidence → Grounded draft → Refund guardrail → Output from LLM

Y → Human review skill → Human review tool → Human review output to customer service

Architecture part 2

Url Validation → Website Scraper Tool → Useful Content Cleaning → Long content chunking → Grounded Summarization → Conciseness Guardrail → Summary Output

Architecture part 3

User Question → Input Validation → Conversation Memory and PDF Retrieval → LangGraph Agent → Decide tools need or not → Optional Calculator/Date Tool → Generate answer → save conversation memory → Output

Part	Trade off	Pros and Cons (trade off)	Potential Failure
1	Hybrid critical detection uses deterministic rules first and DeepSeek semantic checking second. Drafts are restricted to PDF evidence.	Pros: Fixed rules reliably detect explicit mandatory triggers. DeepSeek can recognize unusual paraphrases. Cons: This adds latency and API cost. DeepSeek can still misclassify ambiguous language.	Customer asks a question that is not covered by the internal PDF. Without protection, the model might invent a policy answer. To avoid this, system will block response and request human review to prevent model to guess.
2	Remove webpage noise and divide long content into multiple small chunks instead of sending the complete raw page to DeepSeek.	Pros: Reduces irrelevant HTML and boilerplate. Prevents model context overflow Cons: Cleaning may accidentally remove useful content. Chunking can separate related information.	Invalid, private or local URL - URL validation rejects the request. No useful content after cleaning - No summary is generated, an error is returned
3	LangGraph agent workflow	Pro: Clearly controls each processing step. Supports conversation memory and tool selection. Cons: More complex than a single API call. Requires additional code and dependencies.	Conversation history becomes too large - Only the latest 20 conversation turns are retained. Conversations become mixed between users - Each conversation uses a separate thread ID in SQLite.

Github Link: <https://github.com/Yuki512/Agentic-Architect-Challenge>